# Laboratorio di Architetture Software e Sicurezza Informatica [AAF1569]

## 3 - Software Architectures

SAPIENZA
UNIVERSITÀ DI ROMA

Part of therein contents are based on slides of Proff. Massimo Mecella, John Yang and Ian Sommerville

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# Software Architecture

It's the "**overall organization**" of a software product:

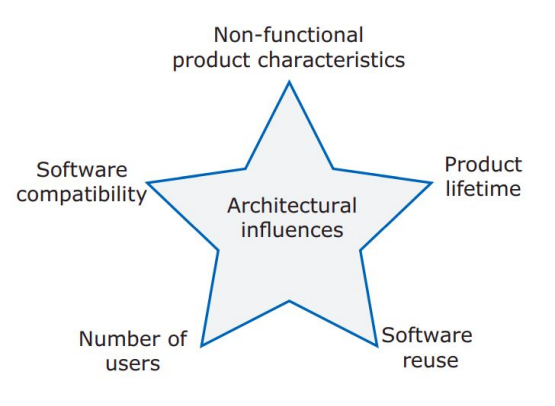
- **software components**
- **server organization**
- **technologies**

**Impacts non-functional** system quality **attributes**:

- responsiveness
- reliability
- availability
- security
- usability
- maintainability
- resilience

**It is impossible to optimize all of the non-functional attributes in the same system**

The software architecture affects the complexity of a product



# Software Architecture

**Trade-offs**:

- **maintainability vs performance**
  - maintainability ⇒ many self-contained parts (easy to replace/enhance)
    ⇒ data exchange among components takes more time (larger comm. overhead)

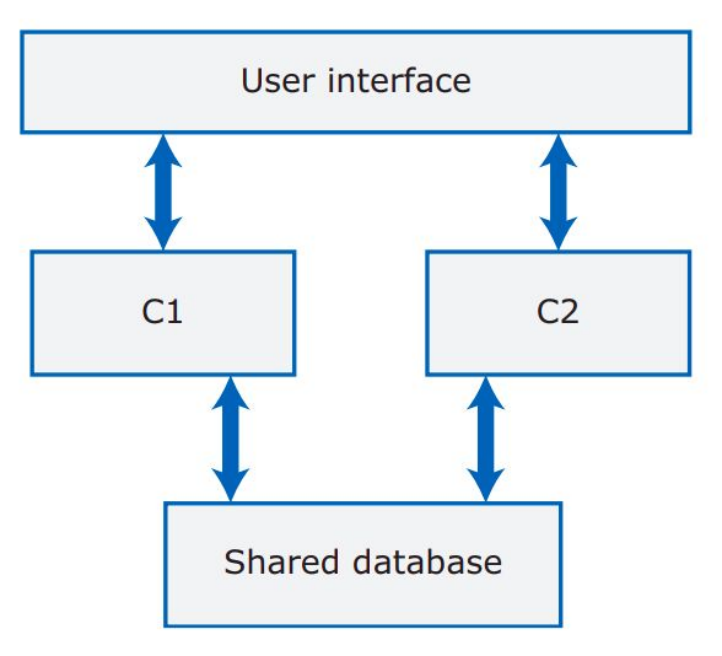
- **security vs usability**
  - multiple security layers ⇒ slower and/or more complex interaction and/or reduced usability

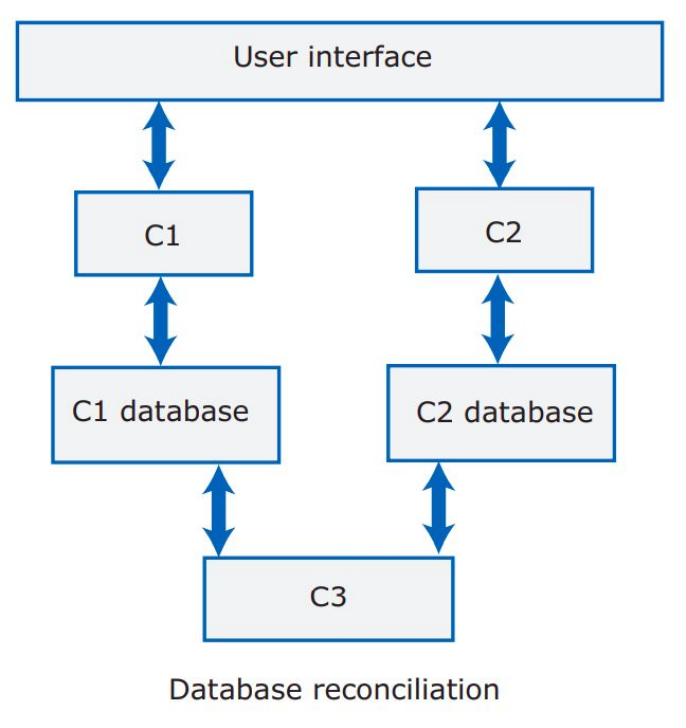
- **availability vs time to market and cost**
  - redundancy ⇒ costs and complexity

Then:

- How to organize the system?
- How the components are distributed and communicate with each other?
- Which technologies are used?

# Architecture Examples



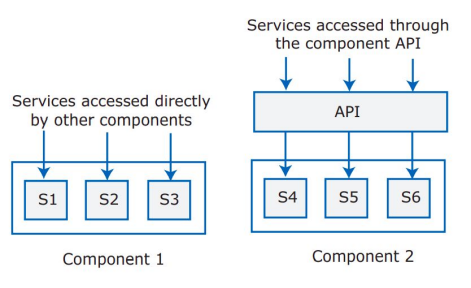
# Software Components: Component, Service and Microservice

A **component** is an element that implements a **coherent set of functionality** or features, i.e. it offers one or more services, that may be used by other components.

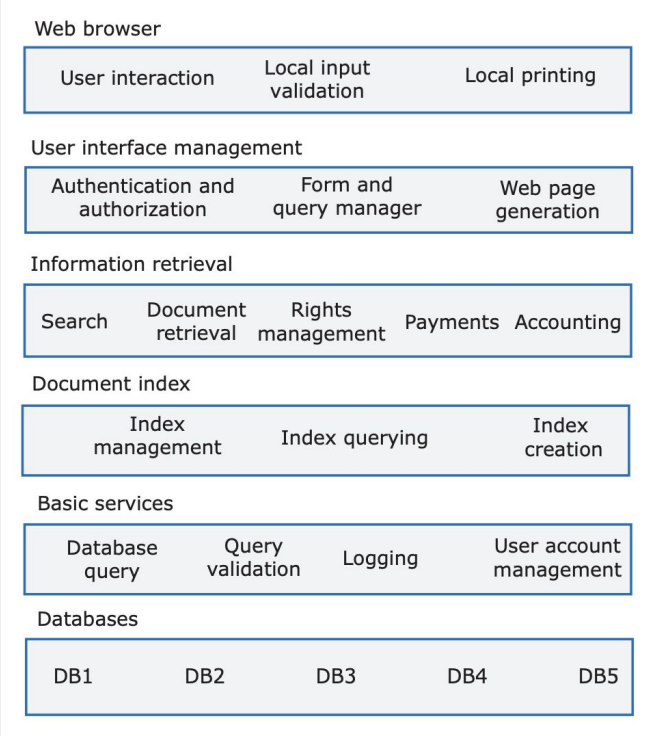
A **service** is a **coherent fragment of functionality**

It can range from a large-scale service (e.g. database) to a **microservice**, which does one **very specific task**.

Services can be accessed directly or through a component API



# Software Components: Component, Service and Microservice - Example



# Why component decomposition?

Components can be:

- **developed in parallel**
- **reused**
- **replaced**
- **distributed**

# Layers

**Layers** are **logical grouping of components**.

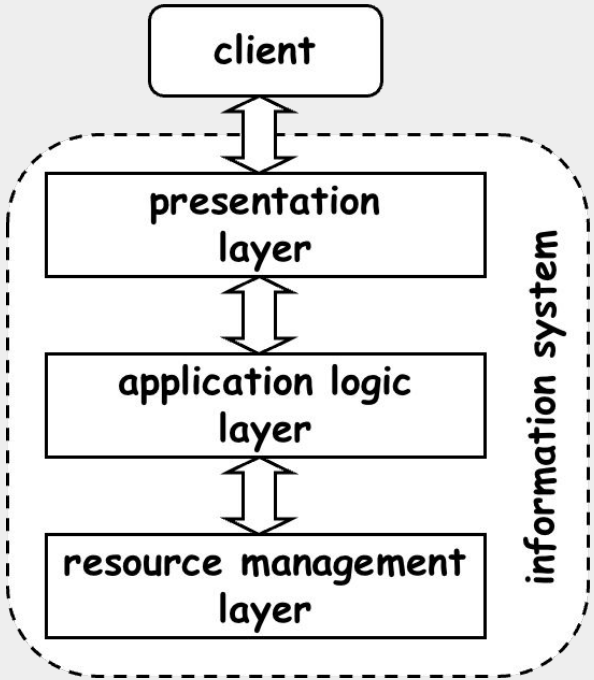
Many software products have a common layered structure

**Presentation**: manage the interactions with the system

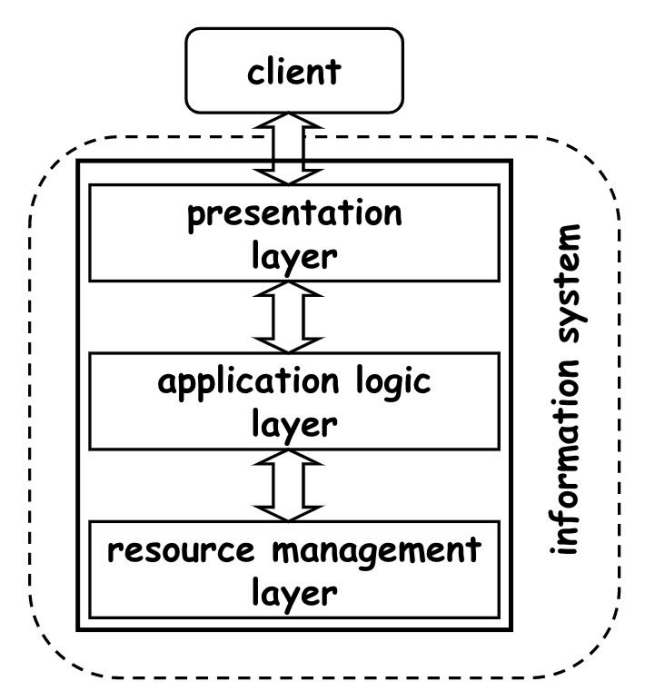
**Application**: what the system does

**Resource management**: deals with the organization of the data

**Tier**: software module implementing a layer



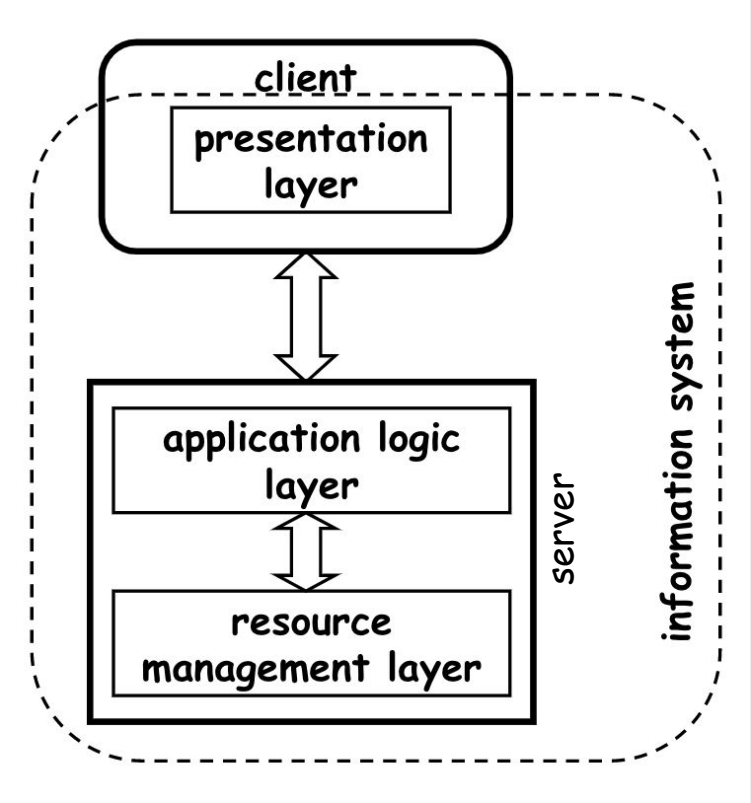
# 1-Tier Architecture




**All is centralized** (single machine)

- low portability
- low maintainability
- harder to update
- no “communication costs”
- easier resource management
- layers can be merged ⇒ better performance

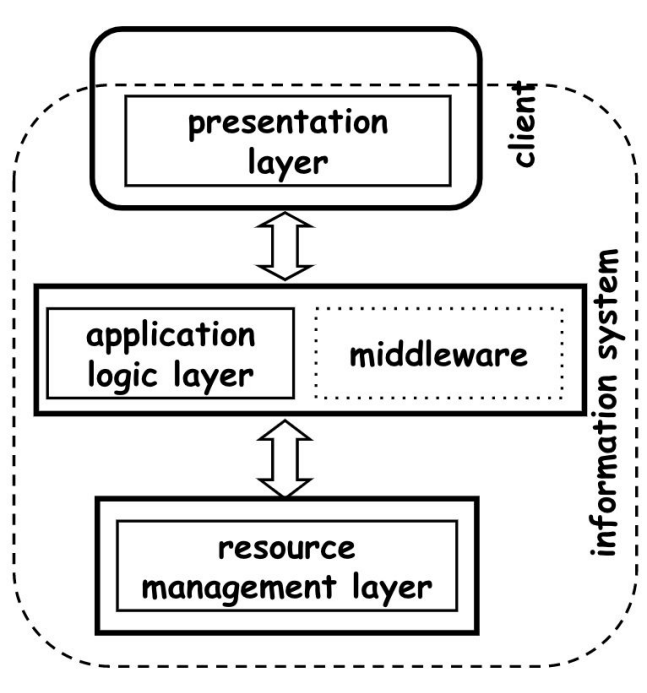
# 2-Tier Architecture




**Clients are independent** of each other: one could have several presentation layers depending on what each client wants to do
⇒ server saves resources

It introduces the concept of **API** (Application Program Interface), an interface to invoke the system from the outside
⇒ it enables integration

# 3-Tier Architecture

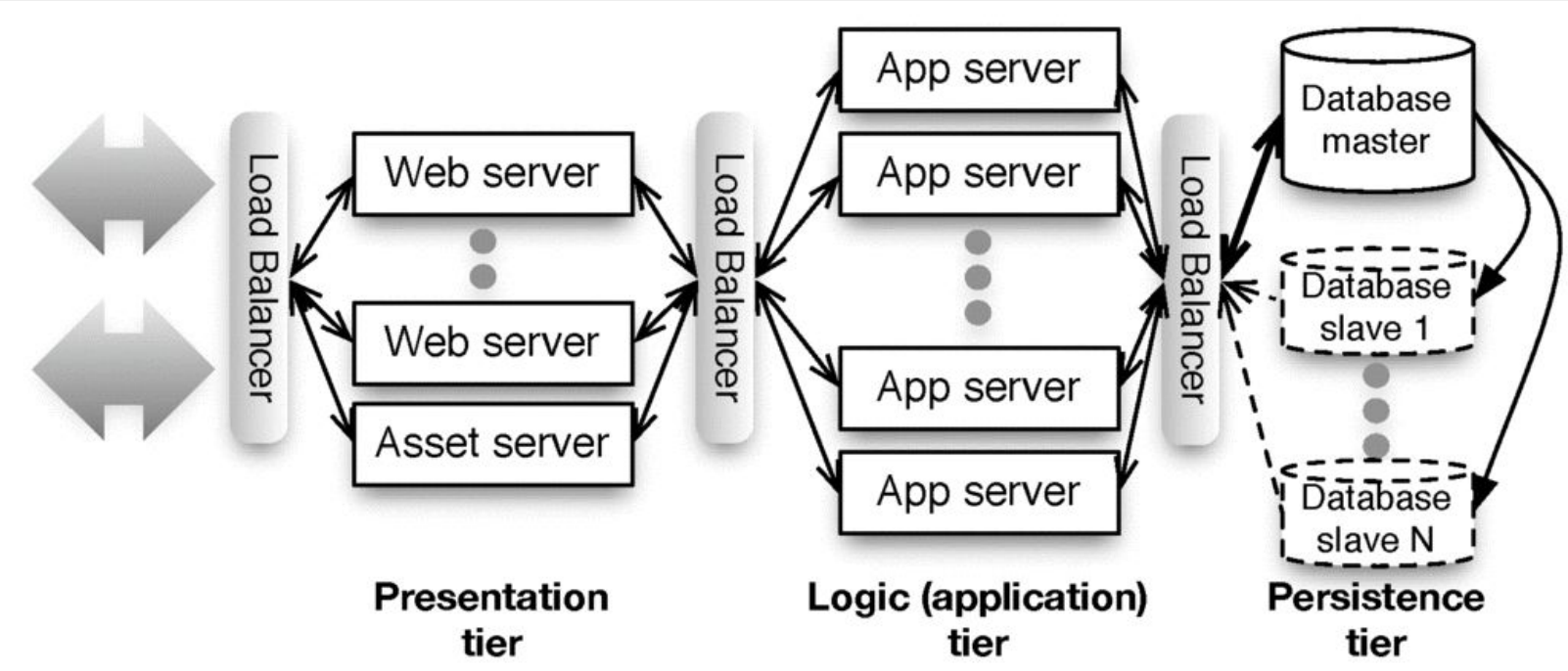



Enables scalability: more servers

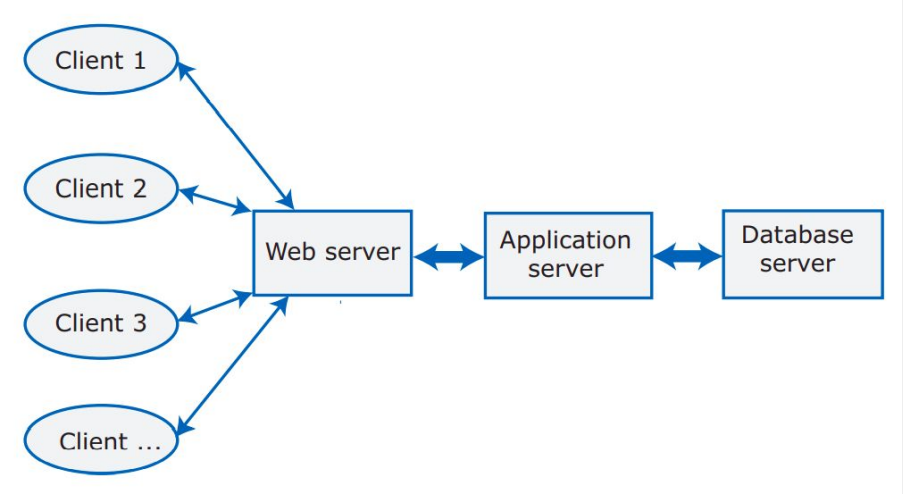
**Middleware:** abstractions & infrastructures supporting the application layer

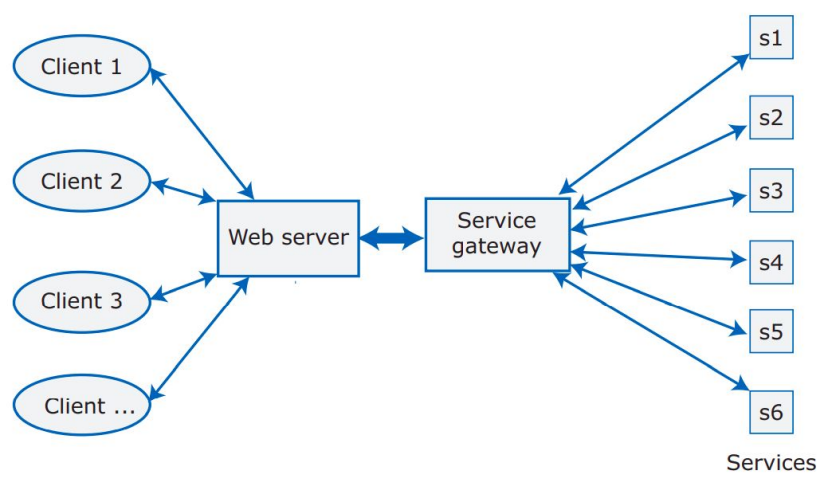
+ more maintainability, easier to update, **more scalability**, can be distributed on different servers
- communication costs between tiers

# 3-Tier Architecture, Scalability



# Distributed Architectures : Multi-tier and Service-oriented



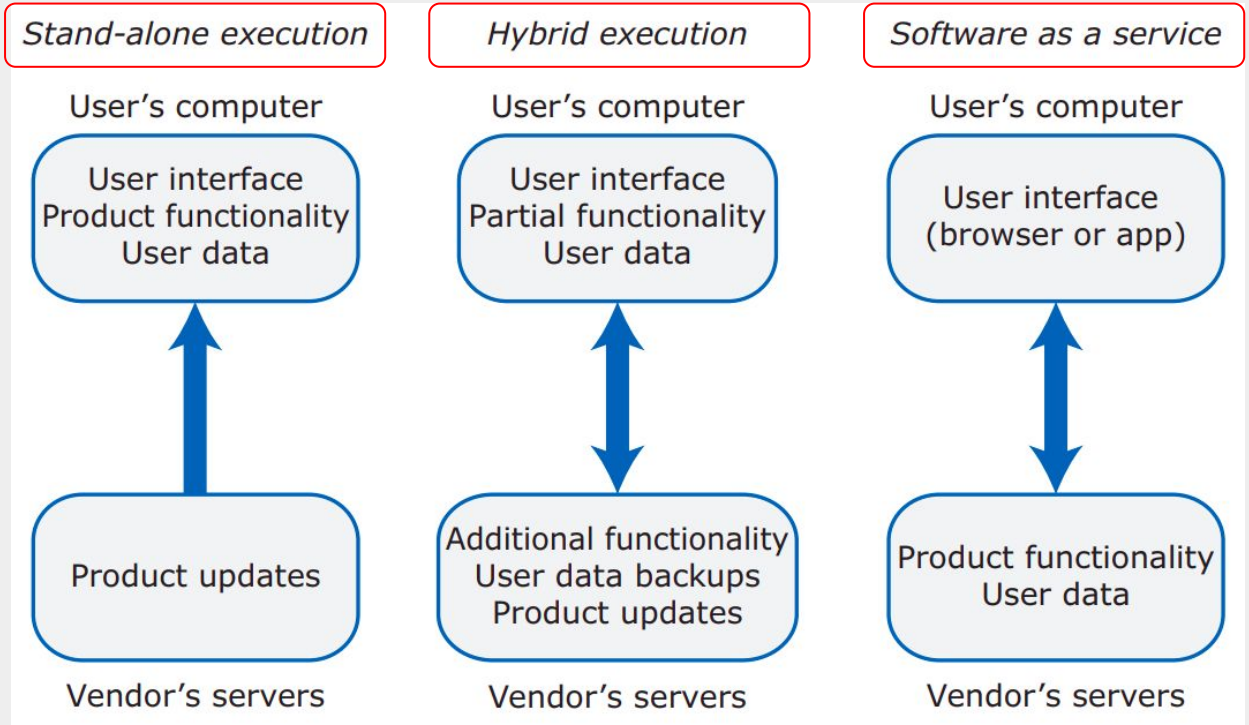




**The choice depends on**:

- Data type and data updates
- Frequency of change
- System execution platform

Services are stateless components => Easier to scale and to update

# Software Execution Models



# SaaS Advantages

1. **No local installation**: removes hardware compatibility requirements
2. **Data is kept within a service** (no backup/data loss concerns for clients)
3. A **single copy** of server software runs in an environment chosen and controlled by developers (e.g., no need to worry about specific OS)
4. **Upgrades** can be handled so as not to interfere with user experience
5. Easier for groups of users to **collectively interact** on same data
6. Remote access is handy for **large or frequently updated data**

Disadvantages?

# Cloud

It is a large number of remote **servers** that are **offered for rent**

Remote servers are “virtual” = implemented in software rather than hardware

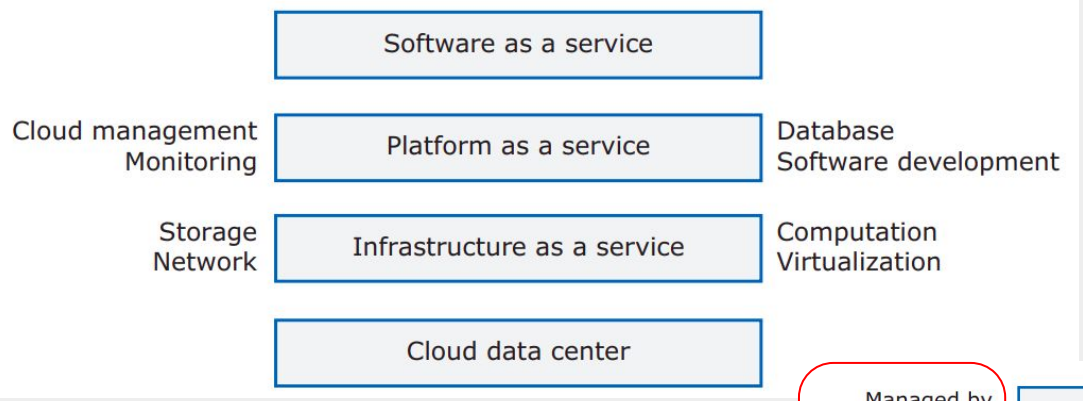
Cloud software features (some): **scalability, elasticity, resilience**

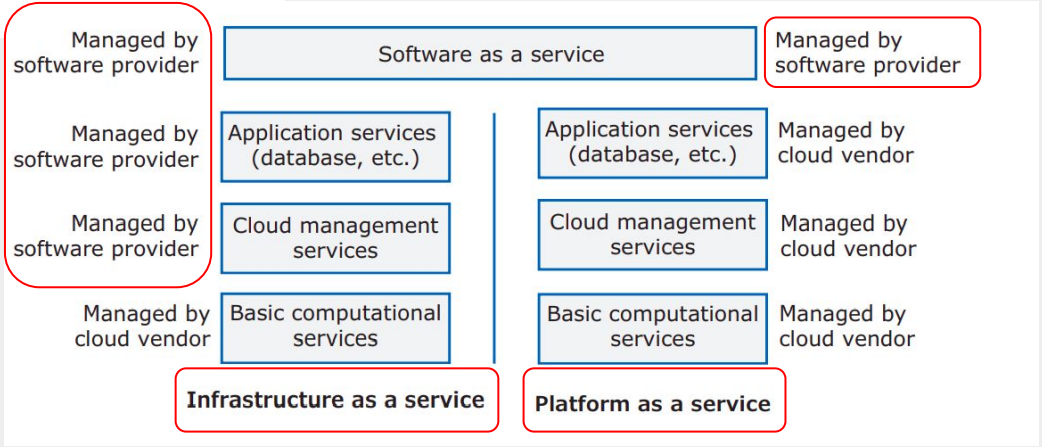
**Infrastructure as a Service (IaaS)**

**Platform as a Service (Paas)**

**Software as a Service (Saas)**

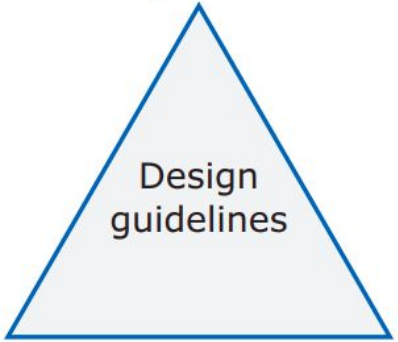
# Cloud


Software as a service
Cloud management
Monitoring
Platform as a service
Database
Software development
Storage
Network
Infrastructure as a service
Computation
Virtualization
Cloud data center


Managed by software provider
Software as a service
Managed by software provider
Managed by software provider
Application services (database, etc.)
Application services (database, etc.)
Managed by cloud vendor
Managed by software provider
Cloud management services
Cloud management services
Managed by cloud vendor
Managed by cloud vendor
Basic computational services
Basic computational services
Managed by cloud vendor
Infrastructure as a service
Platform as a service

# Architectural Design Guidelines

**Separation of concerns**
Organize your architecture into components that focus on a single concern.




**Stable interfaces**
Design component interfaces that are coherent and that change slowly.

**Implement once**
Avoid duplicating functionality at different places in your architecture.